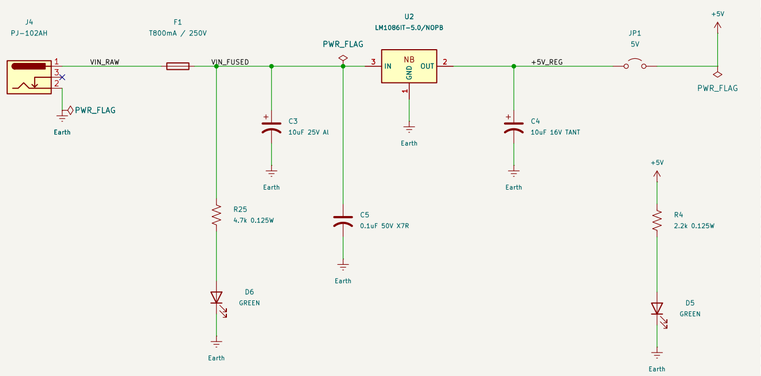
# Fill-A-Bot Main Control Board

## Power Input and 5V Regulator

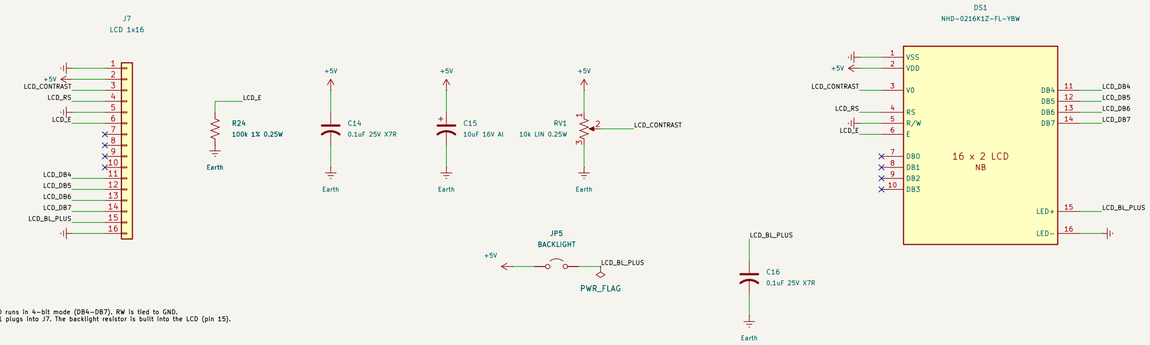



F1 is a 5x20mm fuse in a holder so it can be swapped.

Input is 9–12V DC, center positive.
U2 gets a heat sink (HSE14-38+225P).
D5 and D6 light up when the 5V and 9V rails are on.

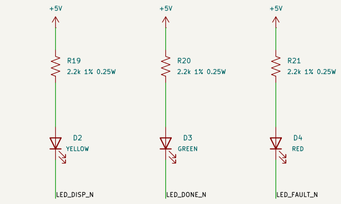



## 16x2 LCD




LCD runs in 4-bit mode (DB4–DB7), RW is tied to GND.
DS1 plugs into J7. The backlight resistor is built into the LCD (pin 15).

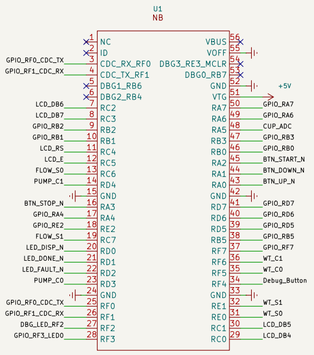
## Status LEDs




## Curiosity Nano, Extra Headers, Debug

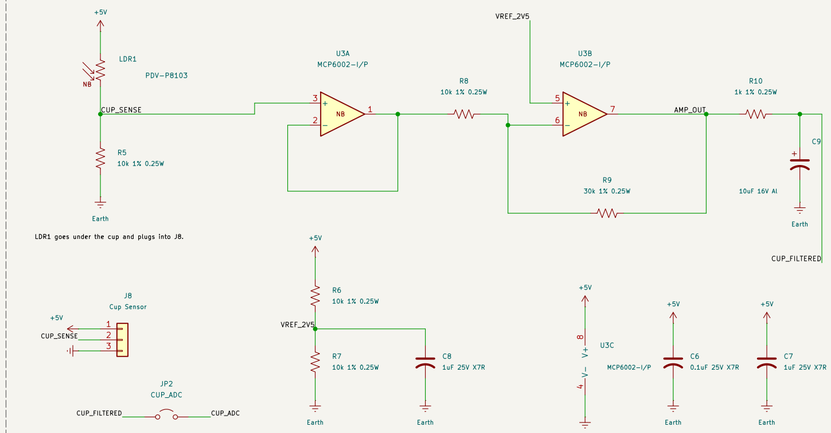
**Ribbon input pull-downs (100k): an unplugged cable reads 00**




Extra headers for spare pins

## Cup Sensor (LDR + Op Amp)



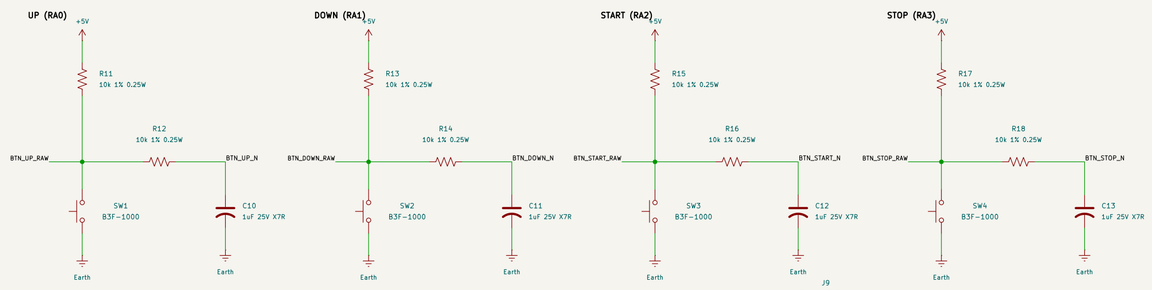

LDR1 goes under the cup and plugs into J8.

U3A buffers the LDR. U3B has a gain of −3 around 2.5V.
R10 and C9 make a low-pass filter (about 16 Hz).

### Test Points



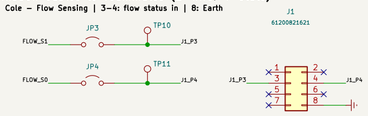

## Buttons




Each button has a 10k pull-up and a 10k/1uF RC debounce.
J9 is for buttons mounted off the board.

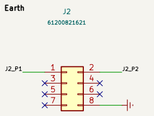
## Ribbon Cables to Teammates (2-wire, 4-state)

**Cole – Flow Sensing | 3–4: flow status in | 8: Earth**



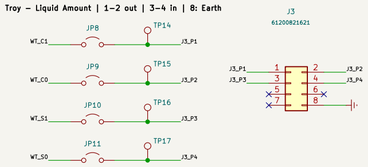

**FLOW_S1:S0 = 00 no flow, 01 flowing,**
**10 not used (no flow), 11 sensor fault**

**Clay – Pump Control | 1–2: pump command out | 8: Earth**




**PUMP_C1:C0 = 00 stop, 01 run,**
**10 and 11 not used (pump stays off)**

**Troy – Liquid Amount | 1–2 out | 3–4 in | 8: Earth**




**WT_C1:C0 = preset 1 to 4 (00, 01, 10, 11)**
**WT_S1:S0 = 00 not ready, 01 below target,**
**10 target reached, 11 error**


**Nikita Bangiyev – EGR 304 Team 201**


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**Title: Fill-A-Bot Main Control Board**

| Size: A2 | Date: 2026-10-25 | **Rev: 3** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |